MMA3001 Project

FEA Surrogate Modelling and Hidden-Property Calibration for a Composite Beam

James Morton – 33853967

Project status: method implemented; current FEA run pending

Abstract

This project develops a reproducible simulation-to-model workflow for predicting the load–deflection response of a composite structure when observations are costly. Dataset A is generated by PyAnsys MAPDL over a regular grid of beam width, laminate thickness and two effective material-property scales. A chosen in-plane/transverse-shear property pair is withheld. Dataset B consists of new MAPDL analyses at that pair, spanning geometries inside and outside Dataset A's width and thickness limits. A mechanics-informed response surface is fitted to Dataset A. One Dataset B response estimates a single common stiffness correction; the remaining Dataset B responses evaluate interpolation and extrapolation separately. The fixed layup, solver setup, grid, response model, calibration objective, and scoring plan are specified below. The current grid workflow is implemented, but its 97 new FEA analyses have not yet been run and no performance result is claimed. This is an FEA-to-FEA methodology demonstration, not physical validation of a landing-gear material model.

1 Problem, aim, and scope

Repeated physical tests may be costly when developing a response model for a composite landing-gear component. The intended future use is to build an FEA-based predictor, correct systematic response error with a small number of observations of the real component, then estimate responses at untested designs. Physical test data are not included in this project. Instead, a controlled FEA-to-FEA exercise uses one FEA dataset with known modelling inputs as the training source and another FEA dataset with deliberately hidden effective properties as the reference relationship.

The immediate benchmark is a straight, prismatic composite cantilever of fixed length, clamped at one end and loaded transversely at its free end. Width and laminate thickness vary. The stack sequence is fixed. The study tests whether the method can learn a response over one property domain, correct it from one response for a withheld property combination, and predict unseen responses within and beyond the training geometry bounds.

The work does not represent curved undercarriage geometry and does not simulate landing impact, contact, large deflection, damage, failure, rate effects, or dynamic response. The term "hidden laminate" refers only to withheld stiffness-scale settings applied to the same 20-ply stack; the ply materials, angles, and sequence are not changed. Results cannot establish the accuracy of the assumed material cards for a physical gear.

2 Finite-element model

2.1 Geometry, element, and laminate

The model is a flat shell mesh on the x–y plane, with the beam axis along global $+X$, width along global $+Y$, and transverse displacement/load along global Z. The planform is rectangular (constant width along the beam); the code sets the taper ratio to one. Beam length is fixed at $L = 0.25\,\mathrm{m}$. The mesh is structured

with 40 divisions along the length and 6 across the width (240 shell elements before any solver-specific internal treatment). The analysis uses the SHELL181 layered-shell element. It is a two-dimensional shell discretization with a layered through-thickness section, not a three-dimensional solid mesh.

Each section has 20 explicit, alternating 0° plies, ordered bottom-to-top: ten carbon 2×2 twill plies (200 gsm) and ten unidirectional E-glass/epoxy plies (100 gsm). Warp is aligned with the beam axis. Nominal cured thickness is 0.20 mm for each carbon ply and 0.15 mm for each glass ply, so the nominal total is 3.50 mm. Three through-thickness integration points are specified per ply. Material cards are orthotropic and include elastic constants, Poisson ratios, shear moduli, and density. Their values are representative modelling inputs assembled from the references recorded in `layup.json`; they are not measured properties of manufactured test coupons. Some transverse properties and cured densities use explicit assumptions. The alternating sequence is unsymmetric and may exhibit extension–bending coupling. The material cards use representative orthotropic constants summarized in Table 1; the complete card, including all Poisson ratios and transverse shear constants, is in `layup.json`. Values are numerical assumptions, not measured project materials.

Table 1: Principal material-card values used at unit property scale. Moduli are in GPa and density in kg/m^3.

Material	E_X	E_Y	E_Z	G_{XY}	G_{XZ}	ν_{XY}	Density
CF 2×2 twill	60.6	61.6	8.3	5.00	3.00	0.10	1724
UD E-glass/epoxy	38.6	8.27	8.27	4.14	4.14	0.26	1111

2.2 Loads, supports, and outputs

Each case is a static, linear-elastic analysis under a total 10 N force in global $-Z$. The force is divided equally among nodes on the free tip edge. All degrees of freedom are fixed at the root edge. The response used for modelling is the magnitude of mean vertical displacement at the tip edge, divided by the applied force:

$$C = \frac{|\bar{u}_{z,\mathrm{tip}}|}{P}, \qquad P = 10\,\mathrm{N}, \tag{1}$$

with units of m/N. Using compliance gives a load-normalized measure for this linear analysis. Each case also records geometry and property scales, calculated section properties, total laminate thickness, mass, signed tip displacement, $|\delta|/L$, load, and mesh divisions.

The model is implemented in `src/run_laminate_shell_sweep.py` and `src/laminate_section.py`; the study driver is `src/run_property_transfer_study.py`. A licensed local MAPDL installation is required for the FEA stages. The solver version, PyMAPDL version, solver output, and run-specific configuration must be retained when the sweep is run. The current 40×6 mesh has not been demonstrated to be mesh-converged for this hidden-property study. The separate convergence workflow should be used as supporting numerical evidence, with its setup and results reported separately.

3 Effective section properties and response model

The 20-ply material cards are used to calculate equivalent laminate bending and transverse-shear stiffnesses. Classical Lamination Theory assembles the A, B, and D matrices from the in-plane ply stiffnesses. The effective bending rigidity per unit width, D_{eff}, is obtained from the inverse of the full ABD matrix with zero membrane resultants and zero non-x bending moments; this retains the unsymmetric stack's extension–bending coupling under the stated assumptions. The approximate beamwise transverse-shear rigidity per unit width is

$$S_{\mathrm{eff}} = \sum_k G_{xz,k} t_k. \tag{2}$$

For width b, the effective beam rigidities are $EI = bD_{\text{eff}}$ and $kGA = \kappa bS_{\text{eff}}$, with $\kappa = 5/6$. These section properties give interpretable predictors and let the training and target configurations be compared through bending and shear contributions.

For each Dataset A case the mechanics basis is

$$x_b = \frac{L^3}{3EI}, \qquad x_s = \frac{L}{kGA}. \tag{3}$$

The fitted model is a non-negative, no-intercept linear regression in these two basis terms:

$$\widehat{C} = c_b x_b + c_s x_s, \qquad c_b \geq 0, \quad c_s \geq 0. \tag{4}$$

The coefficients c_b and c_s are estimated by least squares from Dataset A FEA compliance values. This is a mechanics-informed response surface: it varies smoothly with effective section properties while retaining separate bending and shear contributions. It has two fitted coefficients and uses all Dataset A cases, rather than fitting a high-order polynomial in every raw input. Five-fold shuffled cross-validation on Dataset A is specified to estimate its within-grid predictive performance. The cross-validation score is a model diagnostic; it is not the held-out hidden-property evaluation.

4 Datasets and design of experiments

4.1 Dataset A: training source

Dataset A is a full tensor-product grid with three equally spaced levels on each of four axes:

- width b: 0.030, 0.050, and 0.070 m;
- thickness scale q_t: 0.70, 1.00, and 1.30;
- in-plane stiffness scale q_E: 0.70, 0.90, and 1.10;
- transverse-shear stiffness scale q_G: 0.70, 1.15, and 1.60.

A complete grid would contain $3^4 = 81$ cases. The property pair $(q_E, q_G) = (0.90, 1.15)$ is withheld in full. It would contribute nine width–thickness combinations, so these cases are excluded before FEA and 72 cases are simulated for training. The grid is used because it gives balanced, deterministic coverage of every factor combination and makes deliberate withholding of a whole property pair auditable. It is not a Gaussian quadrature rule. Three levels per axis keep the initial factorial study manageable, but give limited resolution for curvature and interaction effects.

The in-plane scale multiplies E_X, E_Y, and G_{XY} in each material card. The transverse-shear scale multiplies G_{XZ} and G_{YZ}. Thickness scale multiplies each cured ply thickness. Other properties, including density and Poisson ratios, remain fixed. Thus both section rigidity and mass change with thickness; mass is an output for context, not a model target in this response-transfer exercise. The excluded property pair is called a withheld "laminate" setting in the configuration, but is not a different stacking sequence.

4.2 Dataset B: hidden-property target and test set

Dataset B uses the withheld target pair $(q_E, q_G) = (0.90, 1.15)$, with a 5×5 grid of width and thickness:

- width: 0.025, 0.0375, 0.050, 0.0625, and 0.075 m;
- thickness scale: 0.60, 0.80, 1.00, 1.20, and 1.40.

This produces 25 target FEA cases. A case is labelled interpolation only when both width and thickness lie within Dataset A's bounds. Nine Dataset B points satisfy that condition. The other 16 points extend beyond at least one bound and are labelled extrapolation. The central point (width 0.050 m, thickness scale 1.00; case ID 12) is selected as the single calibration observation. It leaves eight interpolation and sixteen extrapolation cases for validation. Dataset B's hidden property scales are used to create the target FEA, but the calibration reads only the observed compliance at the selected case. The hidden settings are retained separately for a post-fit audit and are not inputs to the calibration objective.

Table 2: Planned FEA design and data partition. Counts describe the configured workflow, not completed solver runs.

Set / use	Cases	Contents
Dataset A training	72	3-level grid over width, thickness, and two property scales; the target property pair is excluded.
Dataset B target FEA	25	5×5 width–thickness grid at the withheld property pair.
Calibration	1	Central Dataset B response used to fit a common stiffness correction.
Interpolation validation	8	Remaining Dataset B points inside both Dataset A geometry bounds.
Extrapolation validation	16	Remaining Dataset B points outside at least one Dataset A geometry bound.

5 Training, calibration, and evaluation

5.1 Training Model A

The mechanics basis (x_b, x_s) is computed for each of the 72 Dataset A cases from its width, thickness, and material-property settings. The regression coefficients are chosen to minimize the sum of squared compliance residuals with non-negative coefficients and no intercept. No Dataset B compliance is used to fit Model A. Five-fold cross-validation is run on Dataset A, shuffling folds with the configured seed, to check that the model can represent responses at unseen points drawn from the same training grid.

5.2 One-response calibration

A single compliance cannot separately identify a bending-stiffness error and a shear-stiffness error. The study therefore estimates one scalar s and applies it to both in-plane and transverse-shear moduli of the target layup. Starting from $s = 1$, the bounded range is $0.65 \le s \le 1.35$. For the central calibration case, the selected s minimizes the squared relative residual:

$$\widehat{s} = \underset{0.65 \le s \le 1.35}{\operatorname{argmin}} \sum_{i \in \mathcal{C}} \left(\frac{\widehat{C}_{A,i}(s) - C_{B,i}}{C_{B,i}} \right)^2, \qquad |\mathcal{C}| = 1. \tag{5}$$

Here $C_{B,i}$ is the one observed Dataset B FEA compliance. The same s scales bending- and shear-related material moduli, and the section properties and mechanics basis are recomputed at that candidate scale before prediction. This restricted correction is identifiable from one scalar observation; independent q_E and q_G corrections are not. Therefore, a fitted s is a response-equivalent correction, not a recovery of the two hidden properties.

5.3 Held-out evaluation and why it is separated

Model A and its calibration use no Dataset B validation responses. The calibrated and uncalibrated model predict the remaining 24 cases. Predictions are scored separately for the eight geometry-interpolation cases

and sixteen geometry-extrapolation cases using RMSE, MAE, mean absolute percentage error, maximum absolute percentage error, and R^2 where defined. The calibration case is excluded from these validation scores. Report the fitted correction and compare the nominal and calibrated errors for each region as well as overall. This split tests two different capabilities: local transfer within the sampled geometric range and prediction beyond its boundary. Extrapolation scores should not be interpreted as evidence for transfer to a curved gear, different support, changing length, or real impact conditions.

The workflow uses two simulated datasets to emulate an expensive-observation problem. Because Dataset B is generated by the same FEA code and element setup as Dataset A, the exercise tests design-space coverage, model fitting, parameter correction, and hold-out evaluation under controlled conditions. It does not test FEA model-form discrepancy against physical behaviour. Later work could replace Dataset B observations with physical measurements while keeping the same training, calibration, and evaluation logic, after validating the FEA setup and defining a physical test protocol.

6 Current status and reporting boundary

The current grid workflow is implemented in the repository. Its full configured run comprises 72 Dataset A analyses plus 25 Dataset B analyses, for 97 MAPDL cases. At this report revision, the new result CSVs and model metrics have not been committed. Consequently, this report documents the reproducible method and planned evaluation; it does not claim that the new sweep has run or that calibration improves predictions. After the full workflow is executed, add the fitted correction, cross-validation metrics, regional validation metrics, solver provenance, and run issues.

Earlier quadratic-response-surface, analytical-reference, and mesh-convergence studies in the repository are historical or complementary studies with different configurations. Their metrics must not be presented as results of this 97-case hidden-property grid. The report's numerical results section remains pending until the corresponding grid outputs are generated and reviewed. Before submission, confirm project-scope approval, inspect the model's $|\delta|/L$ values and mesh sensitivity, check the local shell orientation and force/support implementation, and add the student's own AI-use reflection as required by the course brief.

7 Reproducibility

From the repository root, with project Python dependencies, PyMAPDL, and a licensed local MAPDL installation, run:

```
python src/run_property_transfer_study.py --stage all \\
  --config config.hidden_property.example.json --layup layup.json
```

The default CSV paths are `results/property_transfer_grid/dataset_A.csv` and `dataset_B.csv`; the fitted model, predictions, and metrics are written beneath `results/property_transfer_grid/model/`. The analysis can also be run in stages using `--stage dataset-a`, `--stage dataset-b`, and `--stage calibrate`. Calibration requires both CSV datasets but does not launch MAPDL. Record the exact command, MAPDL/PyMAPDL versions, solver log, and configuration used so the run can be reproduced.

8 AI use and student reflection

AI tools assisted with scripting, debugging, drafting, and source discovery. The student is responsible for verifying code, sources, assumptions, and conclusions. Add a personal reflection on one useful contribution, one limitation caught, how outputs were checked, and which decisions were made independently.

References

[1] Ansys, Inc., *SHELL181 Element Documentation*, Ansys Mechanical APDL Element Reference. https://ansyshelp.ansys.com/public/account/secured?returnurl=/Views/Secured/corp/v242/en/ans_elem/Hlp_E_SHELL181.html.

[2] Ansys, Inc., *PyMAPDL secdata API*. https://mapdl.docs.pyansys.com/version/stable/api/_autosummary/ansys.mapdl.core.Mapdl.secdata.html.

[3] J. N. Reddy, *Mechanics of Laminated Composite Plates and Shells: Theory and Analysis*, 2nd ed., CRC Press, 2004.

[4] Gurit, *SE84LV High Performance Prepreg System: Product Data Sheet*, 2024. https://www.revchem.com/document/15pc135966a_tds-pdf/.

[5] University of Cambridge, representative UD glass/epoxy lamina data, https://teaching.eng.cam.ac.uk/system/files/P2B2010QC002.pdf.